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EXPERIMENT 19

Exp No:19 Experiment to understand Logistic Regression for a given data set.

AIM:
To understand Logistic Regression for a given data set.

DESCRIPTION:
Understand the Logistic Regression algorithm for the dataset given.

CODE:

```
import numpy as np

import pandas as pd

df = pd.read_csv('/content/StudentPerformanceFactors.csv')

print(df)

print(df.head())

features = df.iloc[:, [0, 1, 5, 6, 9]].values

label = df.iloc[:, 7].values

print(features)

print(label)


from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.preprocessing import StandardScaler


for i in range(1, 21):
```

```
    x_train, x_test, y_train, y_test = train_test_split(
        features, label, test_size=0.2,
        random_state=i, stratify=label)

    scaler = StandardScaler()
    x_train = scaler.fit_transform(x_train)
    x_test = scaler.transform(x_test)

    model = LogisticRegression(max_iter=1000)
    model.fit(x_train, y_train)

    train_score = model.score(x_train, y_train)
    test_score = model.score(x_test, y_test)

    print("Test {} Train {} Random State {}".format(
        test_score, train_score, i
    ))

x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2,
    random_state=37, stratify=label
)

scaler = StandardScaler()
```

```
x_train = scaler.fit_transform(x_train)

x_test = scaler.transform(x_test)


finalModel = LogisticRegression(max_iter=1000)

finalModel.fit(x_train, y_train)


print(finalModel.score(x_train, y_train))


print(finalModel.score(x_test, y_test))


from sklearn.metrics import classification_report


print(classification_report(
    y_test, finalModel.predict(x_test), zero_division=0
))
```

OUTPUT:

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	\
0	23	84	Low	High	
1	19	64	Low	Medium	
2	24	98	Medium	Medium	
3	29	89	Low	Medium	
4	19	92	Medium	Medium	
...	...	...	...	...	
6602	25	69	High	Medium	
6603	23	76	High	Medium	
6604	20	90	Medium	Low	
6605	10	86	High	High	
6606	15	67	Medium	Low	

	Extracurricular_Activities	Sleep_Hours	Previous_Scores	\
0	No	7	73	
1	No	8	59	
2	Yes	7	91	
3	Yes	8	98	
4	Yes	6	65	
...	...	...	...	
6602	No	7	76	
6603	No	8	81	
6604	Yes	6	65	
6605	Yes	6	91	
6606	Yes	9	94	

```
     Motivation_Level Internet_Access  Tutoring_Sessions
0                 Low             Yes                  0
1                 Low             Yes                  2
2              Medium             Yes                  2
3              Medium             Yes                  1
4              Medium             Yes                  3
...               ...             ...                ...
6602           Medium             Yes                  1
6603           Medium             Yes                  3
6604              Low             Yes                  3
6605             High             Yes                  2
6606           Medium             Yes                  0

[6607 rows x 10 columns]
   Hours_Studied  Attendance Parental_Involvement Access_to_Resources  \
0             23          84                  Low                High
1             19          64                  Low              Medium
2             24          98               Medium              Medium
3             29          89                  Low              Medium
4             19          92               Medium              Medium

  Extracurricular_Activities  Sleep_Hours  Previous_Scores Motivation_Level  \
0                         No            7               73              Low
1                         No            8               59              Low
2                        Yes            7               91           Medium
3                        Yes            8               98           Medium
4                        Yes            6               65           Medium


   Internet_Access  Tutoring_Sessions
0             Yes                  0
1             Yes                  2
2             Yes                  2
3             Yes                  1
4             Yes                  3
[[23 84  7 73  0]
 [19 64  8 59  2]
 [24 98  7 91  2]
 ...
 [20 90  6 65  3]
 [10 86  6 91  2]
 [15 67  9 94  0]]
['Low' 'Low' 'Medium' ... 'Low' 'High' 'Medium']
```

```
Test 0.5068078668683812 Train 0.5072847682119205 Random State 1
Test 0.5068078668683812 Train 0.5072847682119205 Random State 2
Test 0.5068078668683812 Train 0.5072847682119205 Random State 3
Test 0.5068078668683812 Train 0.5072847682119205 Random State 4
Test 0.5068078668683812 Train 0.5072847682119205 Random State 5
Test 0.5068078668683812 Train 0.5072847682119205 Random State 6
Test 0.5068078668683812 Train 0.5072847682119205 Random State 7
Test 0.5068078668683812 Train 0.5072847682119205 Random State 8
Test 0.5068078668683812 Train 0.5072847682119205 Random State 9
Test 0.5068078668683812 Train 0.5072847682119205 Random State 10
Test 0.5068078668683812 Train 0.5072847682119205 Random State 11
Test 0.5068078668683812 Train 0.5072847682119205 Random State 12
Test 0.5068078668683812 Train 0.5072847682119205 Random State 13
Test 0.5068078668683812 Train 0.5072847682119205 Random State 14
Test 0.5068078668683812 Train 0.5072847682119205 Random State 15
Test 0.5068078668683812 Train 0.5072847682119205 Random State 16
Test 0.5068078668683812 Train 0.5072847682119205 Random State 17
Test 0.5068078668683812 Train 0.5072847682119205 Random State 18
Test 0.5068078668683812 Train 0.5072847682119205 Random State 19
Test 0.5068078668683812 Train 0.5072847682119205 Random State 20
0.5072847682119205
0.5068078668683812
```

	precision	recall	f1-score	support
High	0.00	0.00	0.00	264
Low	0.00	0.00	0.00	388
Medium	0.51	1.00	0.67	670
accuracy			0.51	1322
macro avg	0.17	0.33	0.22	1322
weighted avg	0.26	0.51	0.34	1322

RESULT:

Thus, Logistic Regression was implemented successfully to predict student performance using the Student Performance Factors dataset. The model was evaluated using training and testing accuracy, and its performance was assessed using a classification report.